

```

import java.util.ArrayList;

public class Library {

    private ArrayList<Book> books;
    private ArrayList<User> users;

    public Library() {
        books = new ArrayList<>();
        users = new ArrayList<>();
    }

    public void addBook(int id, String title, String author) {
        if (findBookById(id) != null) {
            System.out.println("    [ERROR] A book with ID " + id + " already
exists!");
            return;
        }
        books.add(new Book(id, title, author));
        System.out.println("    [SUCCESS] Book \"" + title + "\" added
successfully.");
    }

    public void viewAllBooks() {
        if (books.isEmpty()) {
            System.out.println("    [INFO] No books available in the
library.");
            return;
        }
        System.out.println("\n    ===== ALL BOOKS =====");
        for (Book book : books) {
            System.out.println(book);
        }
        System.out.println("    Total books: " + books.size());
    }

    public void searchBookByTitle(String keyword) {
        ArrayList<Book> results = new ArrayList<>();

        for (Book book : books) {
            if
(book.getTitle().toLowerCase().contains(keyword.toLowerCase())) {
                results.add(book);
            }
        }

        if (results.isEmpty()) {
            System.out.println("    [INFO] No books found matching \"" +
keyword + "\".");
        } else {
            System.out.println("\n    ===== SEARCH RESULTS =====");
            for (Book book : results) {
                System.out.println(book);
            }
            System.out.println("    Found: " + results.size() + " book(s).");
        }
    }

    public void removeBook(int id) {
        Book book = findBookById(id);

        if (book == null) {
            System.out.println("    [ERROR] No book found with ID " + id +
".");
        }
    }
}

```

```

        return;
    }

    if (book.isIssued()) {
        System.out.println("    [ERROR] Cannot remove book \"" +
book.getTitle()
        + "\" - it is currently issued to User ID: " +
book.getIssuedToUserId() + ".");
        return;
    }

    books.remove(book);
    System.out.println("    [SUCCESS] Book \"" + book.getTitle() + "\"
removed successfully.");
}

    public void addUser(int id, String name) {
        if (findUserById(id) != null) {
            System.out.println("    [ERROR] A user with ID " + id + " already
exists!");
            return;
        }
        users.add(new User(id, name));
        System.out.println("    [SUCCESS] User \"" + name + "\" added
successfully.");
    }

    public void viewAllUsers() {
        if (users.isEmpty()) {
            System.out.println("    [INFO] No users registered in the
library.");
            return;
        }
        System.out.println("\n    ===== ALL USERS =====");
        for (User user : users) {
            System.out.println(user);
        }
        System.out.println("    Total users: " + users.size());
    }

    public void issueBook(int bookId, int userId) {
        Book book = findBookById(bookId);
        User user = findUserById(userId);

        if (book == null) {
            System.out.println("    [ERROR] No book found with ID " + bookId +
".");
            return;
        }

        if (user == null) {
            System.out.println("    [ERROR] No user found with ID " + userId +
".");
            return;
        }

        if (book.isIssued()) {
            System.out.println("    [ERROR] Book \"" + book.getTitle()
+ "\" is already issued to User ID: " +
book.getIssuedToUserId() + ".");
            return;
        }

        book.setIssued(true);
    }

```

```

        book.setIssuedToUserId(userId);

        user.issueBook(bookId);

        System.out.println("    [SUCCESS] Book \"" + book.getTitle()
            + "\" issued to \"" + user.getName() + "\" successfully.");
    }

    public void returnBook(int bookId, int userId) {
        Book book = findBookById(bookId);
        User user = findUserById(userId);

        if (book == null) {
            System.out.println("    [ERROR] No book found with ID " + bookId +
                ".");
            return;
        }

        if (user == null) {
            System.out.println("    [ERROR] No user found with ID " + userId +
                ".");
            return;
        }

        if (!book.isIssued()) {
            System.out.println("    [ERROR] Book \"" + book.getTitle() + "\" is
not currently issued.");
            return;
        }

        if (book.getIssuedToUserId() != userId) {
            System.out.println("    [ERROR] Book \"" + book.getTitle()
                + "\" was not issued to \"" + user.getName() + "\"."
                + " It was issued to User ID: " +
book.getIssuedToUserId() + ".");
            return;
        }

        book.setIssued(false);
        book.setIssuedToUserId(-1);

        user.returnBook(bookId);

        System.out.println("    [SUCCESS] Book \"" + book.getTitle()
            + "\" returned by \"" + user.getName() + "\" successfully.");
    }

    private Book findBookById(int id) {
        for (Book book : books) {
            if (book.getId() == id) {
                return book;
            }
        }
        return null;
    }

    private User findUserById(int id) {
        for (User user : users) {
            if (user.getId() == id) {
                return user;
            }
        }
        return null;
    }
}

```

